Adityaa Saha

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Technical Skills

C, C++, JavaScript, TypeScript, Python **Web:** React, Node.js, Express
Databases: MySQL, MongoDB **Tools:** Git, GitHub **Concepts:** DSA, OOP, COA, DBMS, CN, OS

Projects

32-bit RISC-V CPU Simulator C/C++ — GitHub

- Implemented a cycle-accurate RV32I processor simulator with a 5-stage pipeline (IF, ID, EX, MEM, WB) and a two-pass assembler resolving labels and forward references.
- Added a hazard detection unit with EX/MEM and MEM/WB forwarding paths, load-use interlock stalls, and control-hazard flushing on taken branches.
- Instrumented the pipeline with branch prediction, per-stage tracing, and CPI/stall/flush counters exported as JSON for offline analysis.

LSM Tree Key-Value Store C++ — GitHub

- Built a disk-based key-value store on a Log-Structured Merge Tree, flushing an in-memory MemTable to immutable block-based SSTables with a sparse index for binary-searched lookups.
- Implemented leveled compaction to bound read amplification, merge overlapping key ranges, and reclaim space from tombstoned and overwritten keys.
- Integrated a write-ahead log with replay-on-startup crash recovery and CRC32 checksums to detect corruption on every read path.

DNS Server Implementation C++ — GitHub

- Built a multithreaded DNS server handling RFC 1035 wire-format parsing and serialization over both UDP and TCP, backed by a fixed-size thread pool.
- Served authoritative responses from parsed zone files, forwarded unresolved queries upstream with timeout and retry handling, and cached results with TTL awareness.
- Added per-client rate limiting and domain blocklists, with rotating logs and an HTTP `/metrics` endpoint reporting query volume, cache hit ratio, and error rates.

HTTP Load Balancer C++ — GitHub

- Built an L4/L7 load balancer in C++17 on a custom non-blocking socket layer, with an HTTP/1.1 request parser and keep-alive connection reuse.
- Designed a pluggable scheduling interface supporting round-robin, least-connections, and weighted backend selection strategies.
- Implemented active health checks that eject and reinstate backends on configurable thresholds, alongside runtime reconfiguration and live throughput/latency metrics.

Education

Institute of Technology Bhubaneswar 2023 – 2027
B.Tech in Computer Science and Engineering

Positions of Responsibility

- Core Member, Dramatics Society — The Fourth Wall
- Event Management Team Member, Cultural Fest — Alma Fiesta

Extracurricular Activities

- Member of the IIT Bhubaneswar football team; active in dramatics and stage performances

Achievements

- Secured 2nd Place in Inter-Branch General Championship (Game Development)

Interests

Structures and Algorithms, Systems Programming, Backend Development